

Study and Implementation of Dockerfile and Docker Volume

Objective

The objective of this experiment is to understand the concepts of Dockerfile and Docker Volume, learn their purpose, create a Docker image using a Dockerfile, and use Docker Volumes for persistent data storage.

Introduction

Docker is an open-source platform used to develop, package, and deploy applications inside lightweight containers. Containers include the application along with all its dependencies, ensuring that the application runs consistently across different environments.

Two important Docker concepts are **Dockerfile** and **Docker Volume**.

- A **Dockerfile** is used to automate the creation of Docker images.
- A **Docker Volume** is used to store and manage persistent data independently of the container lifecycle.

These features simplify application deployment and data management.

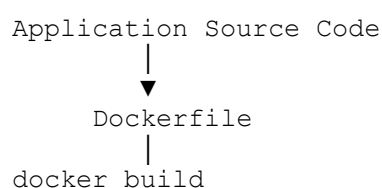
What is a Dockerfile

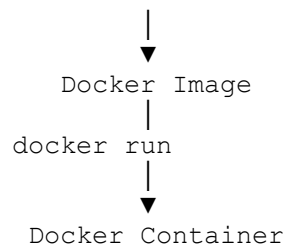
A Dockerfile is a text file containing a sequence of instructions that Docker follows to build an image automatically.

It defines:

- Base image
- Working directory
- Application files
- Dependencies
- Commands to execute
- Startup command

Dockerfile Architecture





Basic Dockerfile Example

```
FROM openjdk:17

WORKDIR /app

COPY . .

RUN javac Hello.java

CMD ["java", "Hello"]
```

Dockerfile Instructions

FROM

Specifies the base image.

Example:

```
FROM openjdk:17
```

WORKDIR

Sets the working directory inside the container.

```
WORKDIR /app
```

COPY

Copies files into the container.

```
COPY . .
```

RUN

Executes commands while building the image.

```
RUN javac Hello.java
```

CMD

Defines the command executed when the container starts.

```
CMD ["java","Hello"]
```

Commands Used for Dockerfile

Build Image

```
docker build -t myapp .
```

View Images

```
docker images
```

Run Container

```
docker run myapp
```

Advantages of Dockerfile

- Automates image creation.
- Provides consistent environments.
- Easy to maintain.
- Portable across systems.
- Supports DevOps and CI/CD.

What is Docker Volume

A Docker Volume is a storage mechanism used to persist data generated by containers. Data stored in a volume remains available even if the container is stopped or deleted.

Volumes are managed by Docker and are stored outside the container's writable layer.

Why Docker Volume is Needed

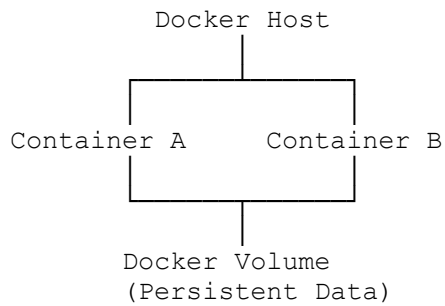
Normally, when a container is deleted, all the data stored inside it is also deleted.

Docker Volumes solve this problem by storing data separately.

Benefits include:

- Persistent storage
- Easy backup
- Data sharing between containers
- Better performance
- Simplified data management

Docker Volume Architecture



Docker Volume Commands

Create a Volume

```
docker volume create myvolume
```

List Volumes

```
docker volume ls
```

Inspect a Volume

```
docker volume inspect myvolume
```

Run a Container Using a Volume

```
docker run -d -v myvolume:/data ubuntu
```

Remove a Volume

```
docker volume rm myvolume
```

Remove Unused Volumes

```
docker volume prune
```

Advantages of Docker Volume

- Stores data permanently.
- Data remains after container deletion.
- Enables data sharing between containers.
- Easy backup and restore.
- Better storage management.
- Improves application reliability.

```
root@ip-172-31-65-154:/home/ubuntu# docker network ls
NETWORK ID          NAME                DRIVER              SCOPE
862a0914951c        bridge              bridge              local
c94817640d85        hospital-network    bridge              local
1e21ee18afa3        host                host                local
1ff0680a350e        none                null                local
root@ip-172-31-65-154:/home/ubuntu# docker network connect hospital-network pandu-container
root@ip-172-31-65-154:/home/ubuntu# docker network disconnect hospital-network pandu-container
root@ip-172-31-65-154:/home/ubuntu# docker network rm hospital-network
hospital-network
root@ip-172-31-65-154:/home/ubuntu# history
1* docker build -t hospital-app
2 docker images
3 docker run -d -p 5000:5000 hospital-app
4 ls
5 docker ps
6 docker stop 23535e5b3fda
7 docker volume rm hospital-volume
8 docker rm 23535e5b3fda
9 docker volume rm hospital-volume
10 docker run -d --name pandu-container -p 5000:5000 hospital-app
11 docker network create hospital-network
12 docker network ls
13 docker network connect hospital-network pandu-container
14 docker network disconnect hospital-network pandu-container
15 docker network rm hospital-network
16 history
root@ip-172-31-65-154:/home/ubuntu#
```

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1

The screenshot shows a web interface for 'CarePlus Hospital'. At the top is a blue navigation bar with the hospital's name and a logo on the left, and links for 'Home', 'Services', 'Doctors', 'Appointment', and 'Contact' on the right. The main content area has a light blue background with the title 'Book Appointment' in bold. Centered on this page is a white form with rounded corners. The form contains five input fields: 'Enter Your Name', 'Enter Email Address', 'Phone Number', 'Select Department' (a dropdown menu), and 'Describe Your Problem' (a text area with a small icon at the bottom right). Below these fields is a blue button labeled 'Submit Appointment'. In the bottom right corner of the page, there is a small blue circular icon with the number '1' inside.

Applications of Docker Volume

- Database storage
- Web server content
- Application logs
- Configuration files
- Shared data between containers
- Backup and recovery

Learning Outcome

During this experiment, the following concepts were learned:

- Understanding Docker and containerization.
- Creating Dockerfiles.
- Building Docker images.
- Running Docker containers.
- Creating and managing Docker Volumes.
- Using persistent storage with containers.
- Managing Docker images, containers, and volumes.

• Conclusion

The experiment successfully demonstrated the creation of a Dockerfile and the use of Docker Volumes. The Dockerfile automated the process of building a Docker image, while Docker Volumes provided persistent storage for container data. Together, these features improve application portability, consistency, and reliability, making Docker an essential tool in modern software development, DevOps, and cloud computing.